Scalability & Future Plan

Date: 11 April 2025

Team ID: SWTID-2026-5240

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 1 Mark

Scalability & Future Plan

This document outlines how the current EduGenie solution can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state. The source document identifies modular architecture, a cloud-based backend, and future enhancements including voice interaction, multilingual support, mobile access, progress tracking, adaptive learning, collaboration, LMS integration, image/PDF input, real-time sync, and smart notifications.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	AI-generated response accuracy can require continuous testing and prompt fine-tuning.	Incorrect or unclear educational responses may affect learning.	High
2	Performance must be maintained across different devices while using cloud-based AI services.	User experience may vary across devices and usage conditions.	Medium
3	Balancing personalization with general usability and addressing data privacy concerns remains challenging.	Personalization and privacy requirements must be handled carefully.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Lightweight FastAPI backend with cloud-based AI models; designed for broad learner access.	Scale the modular FastAPI/cloud architecture for larger user bases and extend access through a mobile application.
2	Data Storage	Current document emphasizes learning content, AI responses, resources, and future progress tracking.	Introduce progress-tracking dashboards and data-driven adaptive learning to organize learner metrics and insights.
3	Performance	Cloud-based backend with a lightweight web interface; local execution is supported.	Optimize prompts and UI, maintain performance across devices, and add offline mobile features for uninterrupted access.
4	Security	The project identifies data privacy as a development challenge.	Strengthen privacy handling as personalization, real-time sync, notifications, image/PDF input, and collaborative features are added.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Voice-based interaction and multilingual support.	Future enhancement	Hands-free learning, improved accessibility, and broader reach across languages.
Phase 2	Mobile application with offline features; progress tracking dashboards.	Future enhancement	Learning anytime/anywhere and clearer visibility of learner progress.
Phase 3	Gamification, adaptive learning paths, group study, teacher/parent dashboards, and LMS integration.	Future enhancement	Higher engagement, more personalized guidance, and collaborative/institutional learning support.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 4	Image/PDF input recognition, real-time sync, smart notifications, and expanded intelligent learning capabilities.	Future enhancement	Snapshot-based doubt solving, resource summarization, continuity, and richer interaction.

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